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Electrosurgical instruments for controlling thermal damage

Abstract

An end effector assembly includes a first jaw member having a first tissue contact surface and a second jaw member having a second tissue contact surface. The second jaw member is pivotally coupled to the first jaw member. The tissue contact surfaces may include raised steps that define an exhaust channel. A longitudinally extending knife slot may be defined in one or both of the tissue contact surfaces. One or more exhaust slots, which may be defined in one or both of the tissue contact surfaces, are in communication with the knife slot. The exhaust channel and the exhaust slots direct fluid away from the jaw members to minimize thermal damage risks associated with tissue sealing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of U.S. patent application Ser. No. 15/742,717, filed on Jan. 8, 2018, now U.S. Pat. No. 11,224,478, which is a U.S. National Stage Application filed under 35 U.S.C. § 371 (a) of International Patent Application Serial No. PCT/CN2015/084306, filed on Jul. 17, 2015, the entire contents of each of which are incorporated by reference herein.

BACKGROUND

(1) The present disclosure relates to electrosurgery, and more particularly to surgical devices, systems, and methods for controlling electrosurgical thermal damage during a tissue sealing process.

(2) Energy-based medical devices such as electrosurgical instruments are typically used in conjunction with energy sources (external energy sources or portable energy sources incorporated into the instruments themselves) to apply and control the application of energy to tissue to thermally treat, e.g., heat, tissue to achieve a desired tissue effect. Electrosurgical forceps, for example, utilize both the mechanical clamping action of the jaw members and the energy provided by the energy source and delivered to tissue to heat tissue grasped between the jaw members to achieve a desired tissue effect, e.g., to seal tissue.

(3) One challenge associated with the sealing process is avoiding unnecessary tissue damage, particularly nerve damage, resulting from thermal spread that may occur during the tissue sealing process.

SUMMARY

(4) Accordingly, the present disclosure is directed to new electrosurgical instruments, systems, and methods that provide efficient and effective tissue sealing by limiting risks associated with thermal spread during tissue sealing. The presently disclosed electrosurgical instruments and/or systems include end effectors with jaw members defining exhaust slots and/or exhaust channels.

Advantageously, these exhaust slots and/or exhaust channels direct fluid (e.g., steam) away from the jaw members of the end effector to enable effective and efficient tissue sealing while limiting undesirable thermal spread.

(5) In one aspect, the present disclosure relates to an end effector assembly having a first jaw member including a first tissue contact surface and a second jaw member including a second tissue contact surface. The second jaw member is pivotally coupled to the first jaw member. One or both of the first and second tissue contact surfaces may be configured to electrically couple to an electrosurgical energy source to receive electrosurgical energy from the electrosurgical energy source. A longitudinally extending knife slot is defined in one or both of the first and second tissue contact surfaces.

(6) An exhaust slot is defined in one or both of the first and second tissue contact surfaces. The exhaust slot extends transverse to the longitudinally extending knife slot and is in communication with the longitudinally extending knife slot. The exhaust slot is configured to direct fluid toward the longitudinally extending knife slot and away from the first and second jaw members. In some embodiments, the exhaust slot is angled relative to the knife slot. The exhaust slot may extend between an open end in communication with the longitudinally extending knife slot to a closed end. In some embodiments, the closed end is distal to the open end. The exhaust slot may be curved between the open and closed ends.

(7) In some embodiments, the first and second jaw members have a tip-biased configuration.

(8) According to another aspect, the present disclosure is directed to a method for controlling fluid during a tissue sealing process. The method involves grasping tissue between a pair of jaw members pivotally coupled together, activating energy to seal tissue grasped between the pair of jaw members, capturing fluid within one or more exhaust slots defined within a tissue contact surface of one of the pair of jaw members as tissue is sealed between the pair of jaw members, and directing the fluid from one or more exhaust slots to a longitudinally extending knife slot of one or both of the pair of jaw members and away from the pair of jaw members.

(9) Directing the fluid from the one or more exhaust slots may include directing the fluid through one or more exhaust slots at an angle relative to the longitudinally extending knife slot and across one or both of the pair of jaw members. The method may involve directing the fluid proximally through the longitudinally extending knife slot.

(10) In yet another aspect, an end effector assembly includes a first jaw member and a second jaw member pivotally coupled to the first jaw member between an open position and a closed position. In the closed position, the first and second jaw members may be movable between an unpinched position and a pinched position. The first and second jaw members may be tip-biased.

(11) A central longitudinal axis extends distally along the first and second jaw members. A first tissue contact surface has a first raised step that extends longitudinally along the first jaw member. A second tissue contact surface has a second raised step that extends longitudinally along the second jaw member. The first and second raised steps are disposed in complementary relation relative to the central longitudinal axis.

(12) An exhaust channel is defined between the first and second raised steps when the first and second jaw members are in the closed position. The exhaust channel is configured to direct fluid away from the first and second jaw members. The exhaust channel may be configured to direct fluid proximally along the first and second jaw members. The first and second tissue contact surfaces may have L-shaped cross-sectional profiles that complement one another to form the exhaust channel. In some embodiments, a proximal end of the exhaust channel is open and a distal end of the exhaust channel is closed. The first and second jaw members may define a closed tip distal to the exhaust channel that blocks the exhaust channel when the first and second jaw members are in the closed position. In some embodiments, the closed tip includes an insulated coating.

(13) In some embodiments, each of the first and second jaw members includes a grasp portion and a beam portion extending from the grasp portion. The grasp portion of the first jaw member supports the first tissue contact surface and the grasp portion of the second jaw member supports the second tissue contact surface. The beam portions of the first and second jaw members define a space therebetween when the first and second jaw members are in the closed position. The exhaust channel may be configured to direct fluid generated during a tissue sealing process through the exhaust channel and proximally toward the space defined between the beam portions to limit thermal spread formed alongside the first and second jaw members during the tissue sealing process.

(14) Other aspects, features, and advantages will be apparent from the description, the drawings, and the claims that follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the disclosure and, together with a general description of the disclosure given above, and the detailed description given below, serve to explain the principles of the disclosure, wherein:

- (2) FIG. 1 is a perspective view of an electrosurgical instrument illustrated in a closed position in accordance with the principles of the present disclosure;
- (3) FIG. 2 is a perspective view of a distal portion of the electrosurgical instrument of FIG. 1 in an open position;
- (4) FIG. 3A is an enlarged, perspective view of a distal portion of the end effector of FIG. 3A with a distal tip thereof shown in phantom for clarity;
- (5) FIG. 3B is an enlarged, cross-sectional view of the end effector of FIG. 3A as taken along section line 3B-3B;
- (6) FIG. 3C is an enlarged, cross-sectional view of the end effector of FIG. 3A as taken along section line 3C-3C;
- (7) FIG. 4A is an enlarged, side view of a distal portion of an end effector of the electrosurgical instrument of FIG. 1, the electrosurgical instrument shown in the closed position with the end effector in an unpinched condition;
- (8) FIG. 4B is an enlarged, side view of the distal portion of the end effector of FIG. 3A with the end effector shown in a pinched condition;
- (9) FIG. 5 is a schematic, top view of the end effector of FIG. 4A illustrating direction of flow through an exhaust channel of the end effector of FIG. 4A;
- (10) FIG. 6 is a top view of a portion of a jaw member of another embodiment of an end effector according to the principles of the present disclosure;
- (11) FIG. 7 is a partial, enlarged, cross-sectional view of a seal plate of the jaw member of FIG. 6 as taken along section line 7-7;
- (12) FIG. 8 is top view of a portion of another jaw member of still another embodiment of an end effector according to the principles of the present disclosure;
- (13) FIG. 9A is a side view of another embodiment of an end effector according to the principles of the present disclosure;
- (14) FIG. 9B is an enlarged, cross-sectional view of the end effector of FIG. 9A as taken along section line 9B-9B;
- (15) FIG. 10 is a side view of one embodiment of an end effector according to the principles of the present disclosure;
- (16) FIG. 11A is a side view of another embodiment of an end effector according to the principles of the present disclosure;
- (17) FIG. 11B is a cross-sectional view of the end effector of FIG. 11A as taken along section line 11B-11B; and
- (18) FIG. 12 is a schematic illustration of a medical work station and operating console in accordance with the present disclosure.

DETAILED DESCRIPTION

- (19) Particular embodiments of the present disclosure are described hereinbelow with reference to the accompanying drawings in which like reference numerals designate identical or corresponding elements in each of the several views. As used herein, the term “distal” refers to that portion of the system, device, and/or component(s) thereof, which is farther from the user, while the term “proximal” refers to that portion of the system, device, and/or component(s) thereof, which is closer to the user. In the following description, well-known functions or constructions are not described in detail to avoid obscuring the present disclosure in unnecessary detail.
- (20) Surgical systems in accordance with the present disclosure can include endoscopic and/or open surgical instruments such as forceps devices, ultrasonic dissection devices, and/or any other suitable surgical devices. Obviously, different electrical and mechanical connections and considerations apply to each particular type of device; however, the aspects and features of the present disclosure remain generally consistent regardless of the particular device used. For a detailed discussion of the construction and operation of example surgical devices, reference may be made to U.S. Patent Application Publication No. 2013/0255063, now U.S. Pat. Nos. 9,265,569,

and/or 8,444,664, the entirety of each of which is incorporated herein by reference.

(21) In the interest of brevity, surgical systems of the present disclosure will only be described herein in connection with an open electrosurgical forceps.

(22) Turning now to FIG. 1, an electrosurgical system, in accordance with the present disclosure, generally referred to as **10**, includes an open electrosurgical forceps **100** that electrically couples to an electrosurgical energy source “E.”

(23) Electrosurgical forceps **100** and/or electrosurgical energy source “E” can be configured to perform monopolar and/or bipolar electrosurgical procedures, including, for example, cutting, coagulation, ablation, and vessel sealing procedures. Electrosurgical energy source “E” may include a plurality of outputs for interfacing with various electrosurgical instruments such as electrosurgical forceps **100** and/or any other suitable energy modality. Electrosurgical energy source “E” may be powered by direct (e.g., DC or a battery) and/or by alternating current. Electrosurgical energy source “E” may include one or more converting devices for converting from DC to AC or vice versa.

(24) Electrosurgical forceps **100** includes a body **110** including a first arm **110a** and a second arm **110b** that are pivotally coupled together by a pin **111**. Embodiments of electrosurgical forceps **100** can include configurations of Allis clamps, Babcock clamps, hemostats, tweezers, forceps, and other suitable graspers.

(25) First and second arms **110a**, **110b** support a handle portion **112** including a pair of handles **112a**, **112b** at a proximal end of first and second arms **110a**, **110b** and an end effector **114** having a pair of jaw members **114a**, **114b** at a distal end of first and second arms **110a**, **110b**. Handle portion **112** is operable to selectively move/pivot the pair of jaw members **114a**, **114b** between open and closed positions for grasping and/or sealing tissue (not shown) therebetween as described in greater detail below. At least a portion of body **110** can include electrically conductive material (e.g., metallic material or the like) that can be an electrically conductive surface. In some embodiments, only portions of body **110** include electrically conductive materials and/or surfaces, such as, for example, one or both of the pair of jaw members **114a**, **114b**. In certain embodiments, the entire body **110** is electrically conductive. Body **110**, or portions thereof, can include insulation material which may be provided as a layer or coating.

(26) One or more supply and/or return lines **116a**, **116b** extend proximally from body **110** and are disposed in electrical communication with the electrically conductive material/surface(s) of body **110**. The one or more supply and return lines **116a**, **116b** are configured to couple the electrically conductive material/surfaces of body **110** to electrosurgical energy source “E” (e.g., via one or more terminals (not shown) of electrosurgical energy source “E”) for electrically communicating with electrosurgical energy source “E.”

(27) For a detailed discussion of the construction and operation of example of similar electrosurgical devices and/or electrosurgical energy sources, reference may be made to U.S. Patent Application Publication No. 2013/0267947, now U.S. Pat. Nos. 9,044,238, and/or 5,026,370, each of which is incorporated herein by reference.

(28) Referring to FIGS. 1 and 2, first and second jaw members **114a**, **114b** are movable relative to one another between a closed position (FIG. 1) and an open position (FIG. 2), as indicated arrow “A.” Each of first and second jaw members **114a**, **114b** includes a beam portion **118a** and a grasp portion **118b** that extends distally from beam portion **118a** to a distal tip **118c**. Distal tips **118c** support insulation material that may be coated thereon.

(29) With reference to FIGS. 3A and 3B, grasp portion **118b** supports a sealing plate **120** having a stepped tissue contact surface **122** that extends longitudinally along longitudinal axis “L-L” (FIG. 1). As seen in FIG. 3B, each stepped tissue contact surface **122** includes a base step **122a** and a raised step **122b** that is vertically and horizontally offset from base step **122a** such that base and raised steps **122a**, **122b** have different heights in the y-direction (e.g., vertical) and different widths in the x-direction (e.g., horizontal). Base step **122a** is wider than raised step **122b** such that stepped

tissue contact surfaces **122** of first and second jaw members **114a**, **114b** are complementary (e.g., reciprocal counterparts of one another) and define an exhaust channel **124** between stepped tissue contact surfaces **122** when first and second jaw members **114a**, **114b** are in the closed position. Exhaust channel **124** may be centrally disposed and/or coaxially aligned with longitudinal axis “L” of end effector **114**.

(30) Referring to FIGS. **4A**, **4B**, and **5**, exhaust channel **124** extends proximally from distal tips **118c** of first and second jaw members **114a**, **114b** to a space **118d** defined between beam portions **118a** of first and second jaw members **114a**, **114b** while first and second jaw members **114a**, **114b** are in the closed position. Exhaust channel **124** functions to direct flow of fluid “F” (e.g., steam) generated during a tissue sealing process of tissue “T” proximally through exhaust channel **124** toward space **118d** to reduce heating of fluid “F” on tissue contact surfaces **122** and limit thermal spread “TS” adjacent first and second jaw members **114a**, **114b**, as described in greater detail below.

(31) Stepped tissue contact surfaces **122** of grasp portions **118b** of first and second jaw members **114a**, **114b** are positioned to engage one another in unpinched (FIG. **4A**) and pinched positions (FIG. **4B**) while first and second jaw members **114a**, **114b** are in the closed position. As indicated by arrows “B” and “C,” beam portions **118a** of first and second jaw members **114a**, **114b** are movable relative to one another to move grasp portions **118b** of first and second jaw members **114a**, **114b** between the unpinched and pinched positions. For instance, first and second jaw members **114a**, **114b** are movable between the unpinched and pinched positions to effectuate tissue sealing of tissue “T” grasped between grasp portions **118b** of first and second jaw members **114a**, **114b** as described in greater detail below.

(32) Grasp portions **118b** of first and second jaw members **114a**, **114b** are tip-biased such that distal portions grasp portions **118b** contact one another before proximal portions of grasp portions **118b** contact one another as first and second jaw members **114a**, **114b** move from the unpinched position to the pinched position.

(33) With reference to FIG. **4A**, in the unpinched position, distal portions **122b** of stepped tissue contact surfaces **122** of first and second jaw members **114a**, **114b** are in contact while proximal portions **122a** of stepped tissue contact surfaces **122** define an angled space or gap **122c** between stepped tissue contact surfaces **122** of first and second jaw members **114a**, **114b**. In this unpinched position, one or more contact forces (e.g., pressure) may be exerted between the first and second jaw members **114a**, **114b**, as illustrated by arrows of unpinched force distribution “UPFD.”

(34) In the pinched position, seen in FIG. **4B**, proximal and distal portions **122a**, **122b** of stepped tissue contact surfaces **122** of first and second jaw members **114a**, **114b** are in contact and exert gripping, squeezing and/or sealing forces (e.g., pressure) between grasp portions **118b** of first and second jaw members **114a**, **114b**, as illustrated by arrows of pinched force distribution “PFD.” The contact and/or gripping/squeezing/sealing forces of unpinched force distribution “UFPD” and/or pinched force distribution “PFD” may change and/or be uniform along at least portions of grasp portions **118b** of first and second jaw members **114a**, **114b**. For example, pressure at a proximal end of stepped tissue contact surfaces **122** (e.g., adjacent angle space **122c**) may be lower than pressure at a distal end of stepped tissue contact surfaces **122** (e.g., adjacent distal tips **118c**). As another example, sealing pressure along grasp portions **118b** of first and second jaw members **114a**, **114b** may be uniform and/or increase toward distal tips **118c**. Sealing pressure exerted along at least portions of grasp portions **118b** while first and second jaw members **114a**, **114b** are in the pinched position may be greater than contact pressure exerted along at least portions of grasp portions **118b** while first and second jaw members **114a**, **114b** are in the unpinched position.

(35) In use, once desired pressure is applied to tissue “T” grasped between first and second jaw members **114a**, **114b** in the pinched position, as illustrated in FIG. **4B**, electrosurgical energy source “E,” is activated to electrically communicate with sealing plate **120** and enable stepped tissue contact surfaces **122** to seal the grasped tissue “T.” During the sealing process, fluid “F,”

such as steam, is released from the grasped tissue “T.” With distal tips **118c** of first and second jaw members **114a**, **114b** providing a block or distal barrier to exhaust channel **124**, effectively closing a distal end of exhaust channel **124**, proximal displacement/exhaustion of fluid “F” may be facilitated as a function of the pinched force distribution “PFD.” For example, with greater force/pressure adjacent distal ends of grasp portions **118b** of first and second jaw members **114a**, **114b**, as compared to proximal ends of grasp portions **118b**, fluid “F” is urged into exhaust channel **124** and displaced, directed and/or exhausted in a proximal direction.

(36) As seen in FIG. 5, exhaust channel **124** directs fluid “F” proximally through exhaust channel **124** and into space **118d**. As fluid “F” passes through exhaust channel **124** and into space **118d**, a temperature of fluid “F” may decrease as a function of time, distance, and/or heat transfer into ambient air. Advantageously, as a result of decreases in temperature of fluid “F,” redirection of fluid “F,” and/or displacement of fluid “F” provided by the features of exhaust channel **124** and/or space **118d** of first and second jaw members **114a**, **114b**, risks of undesirable thermal spread “TS” associated with first and second jaw members **114a**, **114b** is significantly reduced.

(37) Turning now to FIGS. 6 and 7, another embodiment of an end effector includes a first jaw member **200** having a seal plate **210**. A second jaw member (not shown), which opposes first jaw member **200**, may include an identical or substantially identical seal plate **210**. Seal plate **210** includes a tissue contact surface **212** defining a longitudinally extending knife slot **214** that extends along longitudinal axis “L-L” of this embodiment of the end effector. Longitudinally extending knife slot **214** receives a knife (not shown) therein to sever tissue grasped between the jaw members. Transverse exhaust slots **216**, which may be orthogonal to knife slot **214**, are defined within seal plate **210** on one or both opposed sides of knife slot **214**. Exhaust slots **216** extend across seal plate **210** and include a closed end **216a** and an open end **216b**. Open end **216b** is in communication with knife slot **214**. Adjacent exhaust slots **216** may be longitudinally and/or laterally offset from one another.

(38) In use, exhaust slots **216** capture fluid “F” generated during a tissue sealing process and direct the captured fluid “F” into knife slot **214**. Knife slot **214** receives fluid “F” and directs and/or exhausts the captured fluid “F” proximally through knife slot **214**.

(39) With reference to FIG. 8, another embodiment of an end effector includes first jaw member **300** having a seal plate **310**. First jaw member **300** and seal plate **310** are substantially similar to first jaw member **200** and seal plate **210**. A second jaw member (not shown), which opposes first jaw member **300**, may include an identical or substantially identical seal plate **310**.

(40) Seal plate **310** includes a tissue contact surface **312** defining a longitudinally extending knife slot **314** that extends along a centerline “CL-CL” of the end effector. Transverse exhaust slots **316** are defined within seal plate **310** on one or both opposed sides of knife slot **314**. Transverse exhaust slots **316** extend across seal plate **310** and include a closed end **316a** and an open end **316b**. Open end **316b** is in communication with knife slot **314**. Open end **316b** is disposed proximal to closed end **316a**. In some embodiments, open end **316b** is disposed distal to closed end **316a**. One or more exhaust slots **316** may extend transversely to knife slot **314** such that exhaust slot **316** is non-parallel to, and non-orthogonal to, knife slot **314**. One or more co-linear exhaust slots **318** extend longitudinally along seal plate **310** and are co-linear with knife slot **314** and/or centerline “CL-CL” of the end effector. Co-linear exhaust slot **318** includes an open end **318a** that is proximal to closed end **318b**. Exhaust slots **316**, **318** may be longitudinally and/or laterally offset from one another. One or more of jaw member **300**, seal plate **310**, knife slot **314**, transverse exhaust slots **316**, and co-linear exhaust slots **318** may be at least partially curved.

(41) Turning now to FIGS. 9A and 9B, yet another embodiment of an end effector **400** includes a first jaw member **410** and a second jaw member **420**. First jaw member **410** includes a first seal plate **412** and second jaw member **420** includes a second seal plate **422**. A longitudinally extending knife slot **430** is defined by first and second jaw members **410**, **420**. An exhaust channel **440** in communication with knife slot **430** extends longitudinally along longitudinal axis “L-L” on

opposed sides of knife slot **430**. Exhaust channel **440** functions to proximally exhaust fluid “F” alone and/or in conjunction with knife slot **430** during a tissue sealing process similar to other embodiments described herein. Exhaust channel **440** includes a first jaw channel **442** defined with first jaw member **410** and second jaw channel **444** defined within second jaw member **420**. In some embodiments, exhaust channel **440**, first and second jaw channels **442**, **444**, and/or portions thereof, extend only along one side of knife slot **430**. In certain embodiments, exhaust channel **440** includes only one of first and second jaw channels **442**, **444**. In some embodiments, exhaust channel **440**, first and second jaw channels **442**, **444**, and/or portions thereof, extend above and/or below knife slot **430** along longitudinal axis “L-L.”

(42) As seen in FIG. **10**, one embodiment of an end effector **500** includes first and second jaw members **510**, **520** including first and second seal plates **512**, **522**. First and/or second jaw members **510**, **520** define one or more holes **530** therethrough that receive and/or direct fluid “F” from between first and second jaw members **510**, **520** and out through outer surfaces **514**, **524** of one or both of first and second jaw members **510**, **520** during a tissue sealing process. Holes **530** may be in communication with a knife slot (not shown) of first and/or second jaw members **510**, **520**.

(43) Turning now to FIGS. **11A** and **11B**, another embodiment of an end effector **600** includes first and second jaw members **610**, **620** having first and second seal plates **612**, **622**. One or both of at least portions of first and second jaw members **610**, **620** and/or first and second seal plates **612**, **622** are angled laterally outwardly away from a centerline “CL-CL” thereof and/or from the opposing jaw member and/or seal plate **610**, **612**, **620**, **622** to direct fluid “F” from/out of one side of first and second jaw members **610**, **620** during a tissue sealing process.

(44) The various embodiments disclosed herein may also be configured to work with robotic surgical systems and what is commonly referred to as “Telesurgery.” Such systems employ various robotic elements to assist the surgeon and allow remote operation (or partial remote operation) of surgical instrumentation. Various robotic arms, gears, cams, pulleys, electric and mechanical motors, etc. may be employed for this purpose and may be designed with a robotic surgical system to assist the surgeon during the course of an operation or treatment. Such robotic systems may include remotely steerable systems, automatically flexible surgical systems, remotely flexible surgical systems, remotely articulating surgical systems, wireless surgical systems, modular or selectively configurable remotely operated surgical systems, etc.

(45) The robotic surgical systems may be employed with one or more consoles that are next to the operating theater or located in a remote location. In this instance, one team of surgeons or nurses may prep the patient for surgery and configure the robotic surgical system with one or more of the instruments disclosed herein while another surgeon (or group of surgeons) remotely control the instruments via the robotic surgical system. As can be appreciated, a highly skilled surgeon may perform multiple operations in multiple locations without leaving his/her remote console which can be both economically advantageous and a benefit to the patient or a series of patients.

(46) The robotic arms of the surgical system are typically coupled to a pair of master handles by a controller. The handles can be moved by the surgeon to produce a corresponding movement of the working ends of any type of surgical instrument (e.g., end effectors, graspers, knives, scissors, etc.) which may complement the use of one or more of the embodiments described herein. The movement of the master handles may be scaled so that the working ends have a corresponding movement that is different, smaller or larger, than the movement performed by the operating hands of the surgeon. The scale factor or gearing ratio may be adjustable so that the operator can control the resolution of the working ends of the surgical instrument(s).

(47) The master handles may include various sensors to provide feedback to the surgeon relating to various tissue parameters or conditions, e.g., tissue resistance due to manipulation, cutting or otherwise treating, pressure by the instrument onto the tissue, tissue temperature, tissue impedance, etc. As can be appreciated, such sensors provide the surgeon with enhanced tactile feedback

simulating actual operating conditions. The master handles may also include a variety of different actuators for delicate tissue manipulation or treatment further enhancing the surgeon's ability to mimic actual operating conditions.

(48) Referring also to FIG. 12, a medical work station is shown generally as work station **1000** and generally may include a plurality of robot arms **1002**, **1003**; a control device **1004**; and an operating console **1005** coupled with control device **1004**. Operating console **1005** may include a display device **1006**, which may be set up in particular to display three-dimensional images; and manual input devices **1007**, **1008**, by means of which a person (not shown), for example a surgeon, may be able to telemanipulate robot arms **1002**, **1003** in a first operating mode.

(49) Each of the robot arms **1002**, **1003** may include a plurality of members, which are connected through joints, and an attaching device **1009**, **1011**, to which may be attached, for example, a surgical tool “ST” supporting an end effector **1100**, in accordance with any one of several embodiments disclosed herein, as will be described in greater detail below.

(50) Robot arms **1002**, **1003** may be driven by electric drives (not shown) that are connected to control device **1004**. Control device **1004** (e.g., a computer) may be set up to activate the drives, in particular by means of a computer program, in such a way that robot arms **1002**, **1003**, their attaching devices **1009**, **1011** and thus the surgical tool (including end effector **1100**) execute a desired movement according to a movement defined by means of manual input devices **1007**, **1008**. Control device **1004** may also be set up in such a way that it regulates the movement of robot arms **1002**, **1003** and/or of the drives.

(51) Medical work station **1000** may be configured for use on a patient “P” lying on a patient table **1012** to be treated in a minimally invasive manner by means of end effector **1100**. Medical work station **1000** may also include more than two robot arms **1002**, **1003**, the additional robot arms likewise being connected to control device **1004** and being telemanipulatable by means of operating console **1005**. A medical instrument or surgical tool (including an end effector **1100**) may also be attached to the additional robot arm. Medical work station **1000** may include a database **1014**, in particular coupled to with control device **1004**, in which are stored, for example, pre-operative data from patient/living being “P” and/or anatomical atlases.

(52) Persons skilled in the art will understand that the structures and methods specifically described herein and shown in the accompanying figures are non-limiting exemplary embodiments, and that the description, disclosure, and figures should be construed merely as exemplary of particular embodiments. It is to be understood, therefore, that the present disclosure is not limited to the precise embodiments described, and that various other changes and modifications may be effected by one skilled in the art without departing from the scope or spirit of the disclosure. Additionally, the elements and features shown or described in connection with certain embodiments may be combined with the elements and features of certain other embodiments without departing from the scope of the present disclosure, and that such modifications and variations are also included within the scope of the present disclosure. Accordingly, the subject matter of the present disclosure is not limited by what has been particularly shown and described.

Claims

1. An end effector assembly comprising: a first jaw member including a first conductive tissue contact surface; a second jaw member including a second tissue contact surface, the second jaw member pivotally coupled to the first jaw member; a longitudinally extending knife slot defined in the first conductive tissue contact surface; an exhaust slot defined in the first conductive tissue contact surface, the exhaust slot extending transverse to the longitudinally extending knife slot and in communication therewith, the exhaust slot configured to direct fluid toward the longitudinally extending knife slot, and a second exhaust slot defined in the first conductive tissue contact surface.
2. The end effector assembly according to claim 1, wherein the exhaust slot is disposed at a non-

perpendicular angle relative to the longitudinally extending knife slot.

3. The end effector assembly according to claim 1, wherein the exhaust slot includes an open end in communication with the longitudinally extending knife slot and a closed end.

4. The end effector assembly according to claim 3, wherein the exhaust slot is curved between the open end and the closed end.

5. The end effector assembly according to claim 3, wherein the closed end is disposed distally of the open end.

6. The end effector assembly according to claim 1, wherein each of the first jaw member and the second jaw member includes a tip-biased configuration.

7. The end effector assembly according to claim 1, wherein the first conductive tissue contact surface is configured to electrically couple to an electrosurgical energy source to receive electrosurgical energy.

8. The end effector assembly according to claim 1, wherein the exhaust slot is configured to direct fluid proximally along the first jaw member.

9. A jaw member for an end effector assembly, the jaw member comprising: a conductive tissue contact surface defining a perimeter; a longitudinally extending knife slot defined in the conductive tissue contact surface; a first exhaust slot defined in the conductive tissue contact surface, the first exhaust slot extending a first distance on a first lateral side of the longitudinally extending knife slot, transverse to the longitudinally extending knife slot, and disposed in fluid communication with the longitudinally extending knife slot, the first exhaust slot being spaced from the perimeter of the conductive tissue contact surface, and a second exhaust slot defined in the conductive tissue contact surface, wherein the second exhaust slot extends a second distance on a second lateral side of the longitudinally extending knife slot, transverse to the longitudinally extending knife slot, and disposed in fluid communication with the longitudinally extending knife slot, the second exhaust slot being spaced from the perimeter of the conductive tissue contact surface.

10. The jaw member according to claim 9, wherein the second distance is the same as the first distance.

11. The jaw member according to claim 9, wherein the second distance is different than the first distance.

12. The jaw member according to claim 9, wherein the first exhaust slot is longitudinally aligned with the second exhaust slot.

13. The jaw member according to claim 9, wherein the first exhaust slot is longitudinally offset from the second exhaust slot.

14. The jaw member according to claim 9, further comprising a third exhaust slot defined in the conductive tissue contact surface, the third exhaust slot extending a third distance on the first lateral side of the longitudinally extending knife slot, transverse to the longitudinally extending knife slot, and disposed in fluid communication with the longitudinally extending knife slot.

15. The jaw member according to claim 14, wherein the third distance is smaller than the first distance.

16. The jaw member according to claim 15, wherein the third exhaust slot is disposed distally of the first exhaust slot.

17. The jaw member according to claim 14, further comprising a fourth exhaust slot defined in the conductive tissue contact surface, the fourth exhaust slot extending a fourth distance on the second lateral side of the longitudinally extending knife slot, transverse to the longitudinally extending knife slot, and disposed in fluid communication with the longitudinally extending knife slot.

18. The jaw member according to claim 17, wherein the first exhaust slot is longitudinally aligned with the second exhaust slot, the third exhaust slot is longitudinally aligned with the fourth exhaust slot, the first distance is equal to the second distance, the third distance is equal to the fourth distance, the first distance is larger than the third distance, and the third exhaust slot is disposed distally of the first exhaust slot.

